Presenting "DeepFake Detection Based on Discrepancies Between Faces and Their Context"

Introduction

The paper develops a novel approach to detecting manipulated images and videos of people's faces. It focuses on face swapping, where a subject's face is transposed into an entirely different context, with a secondary focus on face reenactment, where a subject's face is kept within its context, but is manipulated. The paper targets a discrepancy common to swapping techniques, as distinguished from techniques that train themselves on the artefacts of specific techniques, thereby emphasizing robustness in its contribution to the literature. The approach's motivation, design, and results are evaluated in this presentation.

Research Questions and Motivations

The paper is built around the question of whether there are discrepancies in manipulated images that can be exploited. The authors' answer – that there are exploitable discrepancies between subjects' faces and their context (e.g. neck, hair, and ears) – raises additional questions whose affirmation constitute the assumption underlying the paper's novel approach: are facial manipulation methods limited to the internal part of the face?; and does the context surrounding the face provide sufficient information to identify subjects?

Face swapping and face reenactment techniques have improved significantly with the advent of deep learning approaches, especially generative adversarial networks (Nirkin et al., 2022), leading to the popular nomenclature "DeepFakes". Humans perform little better than chance at detecting whether a video is a deepfake even when warned that one of the videos they see is a deepfake (Lewis et al., 2023). More than merely deceiving the public, the advent of deepfakes can result in a liar's dividend where genuine information can be discredited more easily as general uncertainty over the veracity of information results from the internalisation of deepfakes among the public (Hancock & Bailenson 2021; Lewis et al. 2023). Technological countermeasures that are robust to rapidly evolving techniques are especially useful to combat misuse, elevating the urgency of this paper's contribution.

Contribution and Results

The authors validate their approach using benchmark datasets FaceForensics++ and Celeb-DF-v2, demonstrating that their method outperforms conventional classifiers. The more challenging Celeb-DF-v2 benchmark demonstrates the limits of the paper's approach. Though it still outperforms other methods, the accuracy of the paper's approach[1] is significantly lower than on the easier FF-DF subset of FaceForensics++ (Table 1). Further testing of the approach on additional face swapping techniques that the authors perform on videos from the FaceForensics++ database, this time setting a threshold to produce binary predictions of whether a face has been manipulated, show more limited performance when additional[2] face-swapping techniques are included. with 75% accuracy overall, the results would likely show worse performance if applied to the more challenging Celeb-DF-v2 data. Nonetheless, the paper's approach (and its variations) is shown to hold significance against face swapping techniques upon which it was not trained, summarised by the paper's Tables 4 and 5 reproduced in tables 3 and 4 below. The results show lesser performance against more recent techniques, suggesting a time-limit to the paper's relevance to the field as techniques advance.

[1] Evaluated using frame-level Area Under the Curve of the Receiver Operator Characteristics

[2] Additional to deepfakes included in the original FaceForensics++ set

Table 1: Reproduction of Table 2 comparing the accuracy paper's approach vs others on benchmark datasets; "Ours" refers to the paper's approach

Methods	FF-DF	Celeb-DF-v2
Two-stream [54]	70.1	53.8
Meso4 [8]	84.7	54.8
MesoInception4 [8]	83.0	53.6
HeadPose [53]	47.3	54.6
FWA [45]	80.1	56.9
DSP-FWA [45]	93.0	64.0
VA-MLP [49]	66.4	55.0
VA-LogReg [49]	78.0	55.1
XceptionNet-raw [5]	**99.7**	48.2
XceptionNet-c23 [5]	**99.7**	65.3
XceptionNet-c40 [5]	95.5	65.5
Multi-task [64]	76.3	54.3
Capsule [51]	96.6	57.5
Ours	**99.7**	**66.0**

Table 2: Reproduction of Table 3 Evaluating Accuracy of approaches giving binary classification of videos once threshold is selected on different manipulation techniques

Methods	DF	F2F	FS	NT	Pristine	Total
Steg. Features [11]	73.6	73.7	68.9	63.3	34.0	51.8
Cozzolino *et al.* [10]	85.4	67.8	73.7	78.0	34.4	55.2
Rahmouni *et al.* [12]	85.4	64.2	56.3	60.0	50.0	58.1
Bayar and Stamm [9]	84.5	73.7	82.5	70.6	46.2	61.6
MesoNet [8]	87.2	56.2	61.1	40.6	72.6	66.0
Xception [5]	96.3	86.8	90.3	80.7	52.4	71.0
Ours	94.5	80.3	84.5	74.0	67.6	**75.0**

Table 3: Reproduction of Table 4 showing accuracy of paper's method and its variants on unseen methods

	3D-based swap			FSGAN		
Methods	Fake	Real	Total	Fake	Real	Total
Face identity difference	47.33	77.66	62.50	34.66	80.50	57.58
Binary XceptionNet [10]	55.38	**97.72**	76.55	24.80	94.68	59.74
Ours (frozen)	52.79	96.44	74.62	34.76	92.46	63.61
Ours (end-to-end)	54.74	97.70	76.22	31.66	**95.38**	63.52
Ours (concat)	55.42	96.54	75.98	41.64	93.30	67.47
Ours	**68.20**	95.10	**81.65**	**47.14**	90.56	**68.85**
Face identity difference	60.20	66.12	63.16	38.96	77.50	58.23
Binary XceptionNet [10]	89.03	81.36	85.20	73.92	64.04	68.98
Ours (frozen)	85.52	86.92	86.22	67.66	76.20	71.93
Ours (end-to-end)	90.77	83.54	87.16	**79.58**	71.40	**75.49**
Ours (concat)	**91.41**	84.34	87.87	71.92	**78.00**	74.96
Ours	90.52	**88.20**	**89.36**	78.72	71.66	75.19

Table 4: Reproduction of Table 5 showing accuracy on unseen image laundering attacks

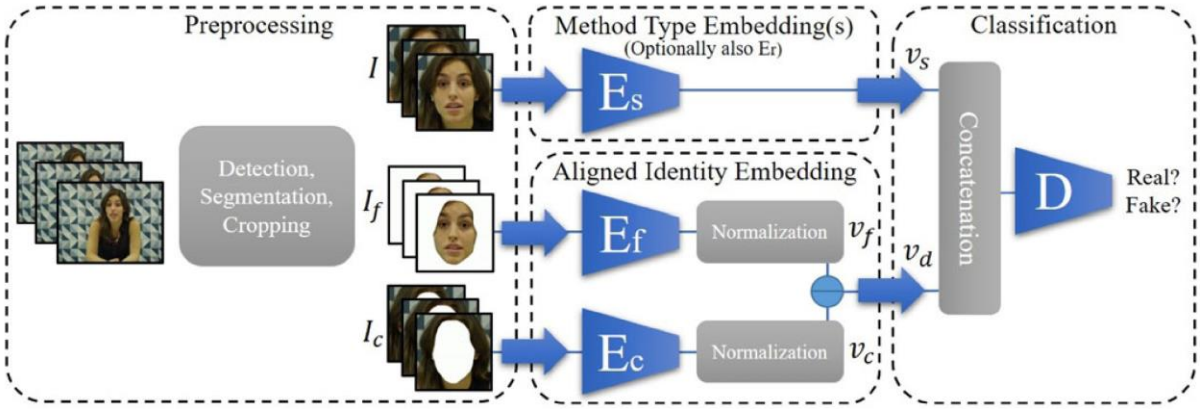
	3D-based swap							FSGAN							FaceShifter						
Methods	RAW	C25	C50	C75	S25	S50	S75	RAW	C25	C50	C75	S25	S50	S75	RAW	C25	C50	C75	S25	S50	S75
Face identity difference	63.16	62.41	63.19	62.73	61.67	62.58	63.02	58.23	57.79	57.32	56.25	56.25	57.72	58.13	55.89	55.92	56.62	55.65	54.19	55.33	55.85
Binary XceptionNet [10]	85.20	84.00	83.07	80.24	75.87	82.09	83.92	68.98	67.68	66.66	63.68	64.77	69.84	70.06	63.62	63.79	62.76	61.90	59.35	63.51	64.43
Ours	**89.36**	**87.49**	**85.79**	**82.21**	**77.51**	**85.46**	**88.06**	**75.19**	**73.60**	**71.96**	**68.61**	**68.07**	**75.03**	**75.85**	**67.64**	**66.96**	**65.85**	**64.28**	**61.47**	**66.66**	**67.52**

Method

The paper's novel approach relies heavily on convolutional neural network architecture, including in its data preprocessing with the use of Dual Shot Face Detector (DSFD) to split the input images on which the approach is trained into subjects' facial region and its complimentary context. DSFD is an effective method that is still used as a benchmark in later work (Zhang et al., 2021). The method proceeds to train two networks based on Xception architecture with cross entropy loss to detect the identities of subjects using just the face identified by DSFD (E_f) and just the complimentary context (E_c). The pseudo-probability vector outputs of E_f and E_c are subtracted from each other to highlight discrepancies in identification, and combined with the penultimate layer of an Xception network directly trained to detect face swapping (E_s). An optional extra Xception trained to detect face reenactment can be added (E_r). These output vectors are concatenated and combined using a classifier (D) with an initial linear layer, followed by batch normalization, ReLU, and a final linear layer – a relatively simple network whose parsimony can act against overfitting. Only the weights of E_s and E_r are fine tuned as D is trained, although the authors test freezing and unfreezing all weights, as well as concatenating rather than subtracting the outputs of E_f and E_c

Though Xception is used in benchmarking later work, it has since been surpassed by more recent convolutional neural networks such as EfficientNetv2 in some applications (Shavit et al., 2022; Tan and Le, 2021), presenting an opportunity for improvement in the paper's method. A paradigm shift outside of convolutional neural networks may also prove fruitful, as vision transformers have emerged at the cutting-edge of deepfake detection. Though computationally more expensive, they outperform configurations of the Xception network (Shah et al., 2022).

Figure 1: Reproduction of figure 4 illustrating the approach's architecture



Critique

Whilst the authors acknowledge their foundational assumption that facial manipulation is limited to the internal part of the face is subject to advancements in face swapping, they fail to consider how their second assumption could fail. Face detection can be made more difficult when subjects wear headgear that occludes their face's context, such as the hijab (Sikandar et al., 2017). The signal given by the facial context in such cases is reduced, undermining the paper's approach and perpetuating the existing bias in deepfake detection (Trinh and Liu, 2021), which is particularly impactful for vulnerable groups.

The authors only touch upon the challenges from more difficult datasets and alternative techniques with the same scope. The majority of the paper's tests are conducted on the relatively easier and occlusion-free FaceForensics++ dataset rather than the more challenging Celeb-DF-v2 dataset, limiting the lessons for improvement that can be drawn from the greater challenge. The Face X-ray method is presented as another method that focuses on a common trait shared by most face swapping techniques, but the paper's results are not benchmarked against this method, precluding a more complete evaluation of the paper's contribution to the leading edge of face swapping detection.

The paper's ad-hoc approach to dealing with face reenactment, for which the paper's key insight of comparing face to context holds lesser relevance, leaves room for improvement. The results do not delineate the approach's performance exclusively on face reenactment and whether the paper's approach results in an improvement over a baseline. Moreover, the approach relies on human judgement on whether to include the E_r network, which may not be reliable. The integration of that decision into the approach's architecture would improve its capacity to generalise onto unseen data whilst reducing arbitrariness in its structure.

References

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